PPGEE2249 Aprendizado de Máquina
Assignment 3
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▶ Question 1

Train and evaluate a Feedforward Neural Network for a multiclass classification task. Split your data into a training set, a validation set (used to define the stopping criterion - e.g., number of epochs), and a test set for the final evaluation of the model. The output layer should use one Softmax neuron per class, and the training objective should be the minimization of cross-entropy loss. Describe your dataset, the network architecture (hidden units, layer dimensions, and number of layers), and discuss the overall performance (data splitting strategy, classification accuracy, and confusion matrix).

- $0 \leq l \leq L$ (layer index)
- $1 \leq t \leq T$ (input index)
- $\mathbf{x}_t \in \mathbb{R}^{d\times 1}$ (input)
- $\mathbf{x}_t^* = [1\ \mathbf{x}_t^\top]^\top \in \mathbb{R}^{(d+1)\times 1}$ (augmented input)
- $\phi : \mathbb{R}^{(d+1)\times 1} \to \mathbb{R}^{(d+1)\times 1}$ (activation function)
- $\mathbf{W}_t^l \in \mathbb{R}^{(d+1)\times(d+1)}$ (layer weights)
- $\mathbf{h}_t^l \in \mathbb{R}^{(d+1)\times 1}$ (layer output)
- $\text{softmax} : \mathbb{R}^{c\times 1} \to \mathbb{R}^{c\times 1}$ s.t.

$$\text{softmax}([u_1\ u_2\ \cdots\ u_c]^\top) = [\exp(u_1)\ \ \exp(u_2)\ \ \cdots\ \ \exp(u_c)]^\top / \sum_k \exp(u_k)$$

- $\mathbf{V}_t \in \mathbb{R}^{c\times(d+1)}$ (output weights)
- $\mathbf{y}_t \in \mathbb{R}^{c\times 1}$ (output)
- $\mathbf{r}_t \in \mathbb{R}^{c\times 1}$ (label)
- L (loss/error function)

$$\mathbf{h}_t^l = \begin{cases} \mathbf{x}_t^*, & l = 0 \\ \phi(\mathbf{W}_t^l \mathbf{h}_t^{l-1}), & l > 0 \end{cases} \tag{1}$$

$$\mathbf{y}_t = \text{softmax}(\mathbf{V}_t \mathbf{h}_t^L) \tag{2}$$

$$L_t(\mathbf{W}, \mathbf{V}) = -\sum_{\tau \leq t} \sum_i r_{\tau i} \log y_{\tau i} \tag{3}$$

Problem: change $\mathbf{W}_t^l$, $\mathbf{V}_t$ to minimize L_t.
Solution: apply the chain rule of the derivative.

$$\begin{aligned}
L_t(\mathbf{W}, \mathbf{V}) &= -\sum_{\tau \leq t} \sum_i r_{\tau i} \log y_{\tau i} && (4)\\
&= -\sum_{\tau \leq t} \sum_i r_{\tau i} \log \left(\text{softmax}\left(\mathbf{v}_t^i \mathbf{h}_t^l\right)\right) && (5)\\
&= -\sum_{\tau \leq t} \sum_i r_{\tau i} \log \left(\text{softmax}\left(\sum_{j=0}^d v_{tij} h_{tLj}\right)\right) && (6)\\
&= -\sum_{\tau \leq t} \sum_i r_{\tau i} \log \left(\text{softmax}\left(\sum_{j=0}^d v_{tij} \phi^{(L)} h_{tLj}\right)\right) && (7)
\end{aligned}$$

► Question 2

Train and evaluate a Decision Tree on the same classification problem chosen in Question 1. Present the resulting tree (depth, rule-based description) and compare its performance with the results obtained in Question 1.

► Question 3

Perform an experiment testing different kernel functions and hyperparameter configurations to obtain the best SVM classifier for a binary classification task, using a dataset of your choice. Don't forget to discuss the results.
